P7. Analyzing Thematic Alignment in Scientific Journals

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Abstract

This project focuses on quantitatively assessing whether articles published in a specific journal align with its stated "Aims & Scope". The core objective is therefore to evaluate wheter the journal's content remains consistent with its declared scope over time. The chosen journal for the demonstration is "Frontiers in Artificial Intelligence".
To achieve this, article abstracts are represented through sentence embeddings generated using a domain-specific pre-trained model based on Sentence-BERT. The Aim & Scope text is encoded using the same model and it is used as ground truth. Cosine similarity is computed between each abstract embedding and the scope embedding to obtain an alignment score.
Descriptive statistics and temporal aggregation are employed to analyze the distribution of alignment scores and to detect potential thematic drift across publication years.
The results provide a quantitative framework for assessing coherence between articles and scope in a scientific journal using semantic similarity measures.

1 Introduction

The continuous growth of scientific production leads to the problem of having systematic methods to evaluate and organize scholarly content efficiently. Academic journals define their thematic boundaries through the Aim & Scope section, which specifies the topics and research directions considered relevant within their editorial framework.

Maintaining coherence between published articles and the declared scope is essential for preserving the journal's identity. Typically this coherence is assessed qualitatively, relying on editorial judgment rather than quantitative evidence.

This project proposes a quantitative framework which measures the semantic alignment between published articles and the journal's Aim & scope. Texts are represented as semantic vectors using transformer-based sentence embeddings. The editorial mission is encoded as a reference vector(our ground truth) within the same semantic

space. Cosine similarity is computed between each article abstract and the reference vector, thus producing an alignment score.

The distribution of similarity scores enables the evaluation of thematic consistency and the analysis of potential thematic drift across publication years. The main goal is to quantify the average alignment and examine the statistical behavior of similarity scores in order to assess editorial coherence over time.

2 Dataset

The dataset consists of a collection of scientific articles obtained from a single academic journal. The chosen journal is Frontiers in Artificial Intelligence. The first step was obtaining the ISSN identifier of the journal and the articles were then extracted using the Crossref API. The chosen range of time spans five years, including all the articles published between January 2021 and December 2025.

Each record includes the following fields:

- title
- abstract
- doi (digital object identifier)

These fields provide a representation of the article content. The corpus consists of 1530 articles, which is enough to produce a meaningful distribution of similarity scores and support statistical analysis.

3 Methodology

The proposed approach is based on a semantic analysis pipeline designed to measure the alignment between scientific articles of the chosen journal and its declared Aim & Scope.

The pipeline is built as follow by four stages:

1. text preprocessing
2. embedding generation
3. similarity computation
4. Temporal aggregation and drift analysis

3.1 Text preprocessing

Before generating embeddings, a data cleaning phase is applied. This initial step includes the normalization of whitespace and elimination of non-printable characters. No aggressive linguistic preprocessing is performed, as transformer-based models preserve semantic information more effectively when operating on minimally altered text. BERT-based models are trained on natural language text and they need the full syntactic and semantic structure of sentences to construct meaningful representations.
So, applying aggressive preprocessing would degrade the quality of the resulting

embeddings.

The final output of this phase is a cleaned textual corpus, where each abstract and the scope are represented as a single string. The corpus and the scope are then used as input for the embedding generation process.

3.2 Embedding generation

The core of the methodology used in this project relies on transforming text into vector representations (embeddings). A pre-trained Sentence-BERT model is used in order to accomplish this, which is **allenai-specter**. This model is fine-tuned on scientific papers using citation context as a supervision signal. Two papers linked by a citation are pushed closer together in the embedding space, while unrelated papers are pushed apart.
Each abstract is encoded into a fixed-dimensional vector. The Aim & Scope text is encoded using the same model, generating an embedding that acts as a ground truth representation of the journal thematic focus.

3.3 Similarity computation

The quantification of the thematic alignment between each article and the declared scope is done by using cosine similarity. This computation technique is applied between the abstract embeddings and the scope reference vector.
Let $\mathbf{a}_i$ be $\boldsymbol{i}$-th article and let $\mathbf{e}_{ai}$ and $\mathbf{e}_{scope}$ be the corresponding embeddings vector of the article and the journal scope. The following formula is the actual cosine similarity computed for each article involved.

$$\text{score}(a_i) = \frac{\mathbf{e}_{a_i} \cdot \mathbf{e}_{scope}}{\|\mathbf{e}_{a_i}\| \cdot \|\mathbf{e}_{scope}\|}$$

Cosine similarity is the cosine of the angle between the vectors; it is the dot product of the vectors divided by the product of their lengths. A value close to 1 indicates a high alignment score with the scope; a value close to 0 indicates thematic distance. Negative values are theoretically possible but unlikely in this context given the shared domain.

3.4 Temporal aggregation and drift analysis

Once a score is assigned to each article, the distribution of alignment scores is analyzed both globally and longitudinally. Per-year aggregation is computed by grouping articles by publication year and calculating the mean and standard deviation of scores within each group. This temporal view allows the detection of potential thematic drift, which is a systematic shift in the average alignment over time, as well as changes in score variance, which may reflect increasing editorial heterogeneity.

4 Results

The global distribution of alignment across all articles tell us that the majority of publications has a high semantic alignment to the declared scope of the journal Frontiers in Artificial Intelligence.
As we can see in the tables below, the global mean is over 0.7 and the standard deviation has a value near to zero, which tell us that there's not a much dispersion around the mean. This result is confirmed even if we look at the values filtered by year.

Table 1 Descriptive statistics of alignment score globally

Metric	Value
Mean	0.727
Standard deviation	0.065
Minimum	0.516
Maximum	0.907

Table 2 Descriptive statistics of alignment scores by year

Year	Mean	Std. Dev.	Articles per year
2021	0.7079	0.0648	192
2022	0.7180	0.0611	276
2023	0.7326	0.0591	216
2024	0.7286	0.0689	292
2025	0.7355	0.0661	554

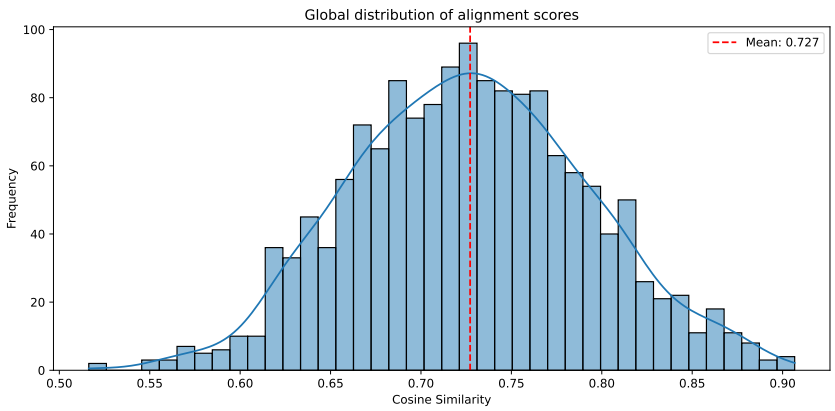

Fig. 1 Global distribution of alignment scores across the dataset.

Related to temporal analysis, the data provides insight about a possible thematic drift over time.
The mean value score increases slightly from 0.7079 in 2021 to 0.7355 in 2025, showing a slight and gradual increase in alignment to the Journal's scope over time. The standard deviation remains relatively stable year by year (0.0591 - 0.0689), with a slight increase in variance in more recent years.
This result suggests this Journal maintains a stable editorial coherence, with no single year exhibiting unusually high variance in alignment.
These obtained data are limited to the selected 2021-2025 time window. A broader analysis would be required to assess whether this pattern holds across older and earlier publication years.
The figure 2 below describes how much the alignment score takes distance from mean every year. It gives a comparison year by year showing dispersion differences.

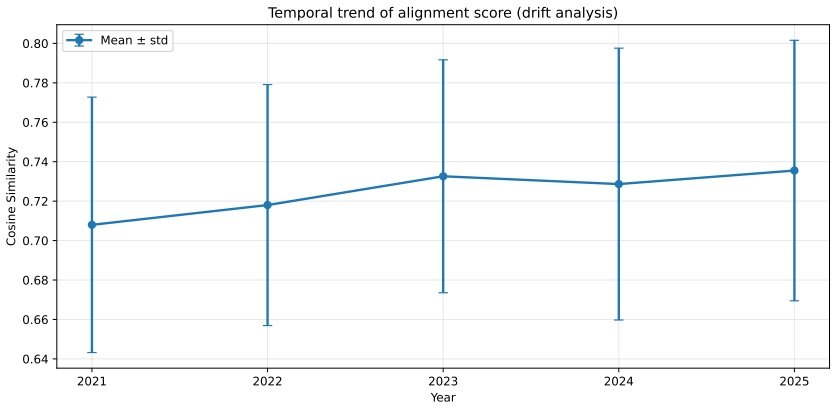

Fig. 2 Temporal trend of alignment score from january 2021 to december 2025.

Extremely low scores represent articles that are less aligned with the journal scope, while high scores close to 0.9 indicate very strong semantic alignment. Overall the analysis confirms that the journal consistently publishes articles coherent with its declared Aim & Scope, with a small number of publications deviating from this focus. Here below it is possible to take a look at the 10 best aligned articles and the 10 worst aligned articles.

Title	Year	Score
Ethics dumping in artificial intelligence	2024	0.907
Specific challenges posed by artificial intelligence in research ethics	2023	0.902
AI in humanitarian healthcare: a game changer for crisis response	2025	0.900
What Makes Artificial Intelligence Exceptional in Health Technology Assessment?	2021	0.898
Knowledge, attitude, and practice of artificial intelligence among doctors and medical students in Syria: A cross-sectional online survey	2022	0.891
Ethical-legal implications of AI-powered healthcare in critical perspective	2025	0.890
The imperative of diversity and equity for the adoption of responsible AI in healthcare	2025	0.887
Perceptions of health data commodification in AI-driven healthcare systems in Saudi Arabia	2025	0.884
Using ChatGPT to navigate ambivalent and contradictory research findings on artificial intelligence	2023	0.881

Table 3 Top 10 articles with highest alignment to the journal's scope

Title	Year	Score
DCM2Net: an improved face recognition model for panoramic stereoscopic videos	2024	0.516
Conformational Changes of the Receptor Binding Domain of SARS-CoV-2 Spike Protein and Prediction of a B-Cell Antigenic Epitope Using Structural Data	2021	0.525
Convolutional Neural Network-Based Technique for Gaze Estimation on Mobile Devices	2022	0.551
Clinical Enhancement in AI-Based Post-processed Fast-Scan Low-Dose CBCT for Head and Neck Adaptive Radiotherapy	2021	0.554
FLA-UNet: feature-location attention U-Net for foveal avascular zone segmentation in OCTA images	2025	0.555
BLoss-DDNet: bending loss and dual-task decoding network for overlapping cell nucleus segmentation of cervical clinical LBC images	2025	0.557
Image restoration in frequency space using complex-valued CNNs	2024	0.562
Deep learning for steganalysis: evaluating model robustness against image transformations	2025	0.562
Isolated single sound lip-reading using a frame-based camera and event-based camera	2023	0.569
NeuroSim Simulator for Compute-in-Memory Hardware Accelerator: Validation and Benchmark	2021	0.569

Table 4 Top 10 articles with lowest alignment to the journal's scope

5 Conclusion

This project introduced a quantitative framework to assess the alignment between scientific publications and the declared scope of a journal. By using semantic similarity techniques, each article was mapped to a continuous alignment score, enabling a systematic and reproducible evaluation process.
In this case study, results showed that Frontiers in Artificial Intelligence maintains a high thematic coherence across the 2021 - 2025 period. The global distribution of scores is concentrated around the mean with limited dispersion. This means most publications are consistently aligned with the declared scope, while a small subset deviates from it.
The main contribution of this project is the introduction of a normalized metric that enables comparison across time, by formalizing the scope alignment as a measurable quantity.

Declarations

Ai Usage Disclaimer: parts of this project have been realized with the assistance of ChatGPT(model GPT-5.3) and Claude(model Sonnet 4.6) for the following types of tasks:

- Reviewing and drafting text during the writing of this report.
- Collecting ideas about results analysis.
- Generation of Python code to support the development.

Every content generated by AI has been reviewed and tested by a human.

References

[1] N. Reimers and I. Gurevych, *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*, In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 3982–3992, 2019.

[2] S. Picascia, S. Montanelli, S. Salini, S., & Verzillo, S. (2025). *The Atlas of Data Science Research*, IEEE Access, 2025.